MODULE 1: CHEMICAL THERMODYNAMICS AND KINETICS

(CO: 1)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 9 Topics): (1) Introduction terminologies, (2) Zeroth law - First law of thermodynamics, Enthalpy and heat capacity, (3) Applications of first law - Second law of thermodynamics, (4) Carnot cycle, (5) Third law of thermodynamics, Gibbs-Helmholtz equation, (6) Chemical kinetics: Rate equation, Order of reaction, Integral rate equations, Half-life time, (7) Collision theory of reaction rate, Effect of temperature and catalysts, (8) Worked Numericals - Part 1, and (9) Worked Numericals - Part 2.

1 VIT FAT Exam Blueprint & Master Formula Sheet

In VIT Final Assessment Tests (BACHY101 FAT), Module 1 carries **15 to 20 marks**. Typically, one 10-mark question focuses on Chemical Thermodynamics (Carnot cycle efficiency, Gibbs-Helmholtz derivation, or enthalpy/entropy calculations) and another 10-mark question appears on Chemical Kinetics (integral rate law derivations, half-life problems, or Arrhenius activation energy determinations).

Concept / Law	Governing Equation / Formula	VIT Exam Key Insight
First Law of Thermodynamics	$\Delta U = q + w = q - P\Delta V$	Work of expansion: $w = -\int P_{\text{ext}} dV$ (IUPAC sign convention).
Heat Capacity Relation	$C_p - C_v = R = nR$ (ideal gas)	$C_p = \left(\partial \frac{H}{\partial} T\right)_P$ and $C_v = \left(\partial \frac{U}{\partial} T\right)_V$.
Carnot Engine Efficiency	$\eta = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{T_H - T_C}{T_H} = 1 - \left(\frac{T_C}{T_H}\right)$	Temperatures T_H, T_C MUST strictly be converted to Kelvin!
Gibbs Free Energy	$\Delta G = \Delta H - T\Delta S$	Spontaneous if $\Delta G < 0$, at equilibrium if $\Delta G = 0$.
Gibbs-Helmholtz Equation	$\Delta G = \Delta H + T\left[\frac{\partial(\Delta G)}{\partial T}\right]_P$	Alternatively: $\left[\partial \frac{\Delta \frac{G}{T}}{\partial} T\right]_P = -\Delta \frac{H}{T^2}$.
First-Order Rate Law	$k = \frac{2.303}{t} \log_{10}\left(\frac{a}{a-x}\right)$	Half life $t_{\frac{1}{2}} = \frac{0.693}{k}$ is completely independent of initial conc a.
Second-Order Rate Law	$k = \frac{1}{t}\left(\frac{x}{a(a-x)}\right)$	Half life $t_{\frac{1}{2}} = \frac{1}{ka}$ is inversely proportional to initial conc a.
Arrhenius Equation	$k = Ae^{-\frac{E_a}{RT}} \Rightarrow \ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$	$R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$; activation energy E_a is strictly positive.

2 Chemical Thermodynamics: Laws, Carnot Engine & Free Energy

2.1 Thermodynamic Systems & The First Law

Thermodynamic systems are classified into open (exchanges matter and energy), closed (exchanges energy only), and isolated (exchanges neither matter nor energy). The **Zeroth Law** establishes thermal equilibrium and the empirical definition of temperature: if body A is in thermal equilibrium with B, and B with C, then A and C are in mutual thermal equilibrium.

The **First Law of Thermodynamics** represents the principle of conservation of energy:

$$\Delta U = q + w$$

where U is internal energy (a state function), q is heat absorbed by the system, and w is work performed on the system. For a quasi-static reversible expansion of an ideal gas against external pressure P:

$$dw = -PdV \Rightarrow dU = dq - PdV$$

At constant pressure, enthalpy H is defined as $H = U + PV$. Differentiating at constant pressure gives $dH = dU + PdV = q_p$. Hence the heat transferred at constant pressure equals the enthalpy change ΔH.

2.2 Heat Capacities & Adiabatic Expansion

The molar heat capacities are defined as:

$$C_v = \left(\partial \frac{U}{\partial} T\right)_V \quad \text{and} \quad C_p = \left(\partial \frac{H}{\partial} T\right)_P$$

For an ideal gas where $H = U + RT$:

$$C_p - C_v = \frac{dH}{dT} - \frac{dU}{dT} = R$$

For a reversible adiabatic process ($dq = 0$):

$$dU = dw \Rightarrow C_v dT = -PdV = -\frac{RT}{V} dV \Rightarrow C_v \frac{dT}{T} = -R \frac{dV}{V}$$

Integrating gives the classic Poisson adiabatic relations:

$$PV^\gamma = \text{constant}, \quad TV^{\gamma-1} = \text{constant}, \quad T^\gamma P^{1-\gamma} = \text{constant}$$

where the adiabatic index $\gamma = \frac{C_p}{C_v}$.

2.3 The Carnot Cycle & The Second Law

A Carnot heat engine operates between a hot reservoir at T_H and a cold sink at T_C via four reversible steps:

1. **Isothermal Expansion ($1 \to 2$ at T_H):** Heat absorbed $q_H = RT_H \ln\left(\frac{V_2}{V_1}\right)$. Internal energy change $\Delta U_1 = 0$.
2. **Adiabatic Expansion ($2 \to 3$ from T_H to T_C):** Heat $q = 0$. Work done $w_2 = C_v(T_C - T_H)$.
3. **Isothermal Compression ($3 \to 4$ at T_C):** Heat rejected $q_C = RT_C \ln\left(\frac{V_4}{V_3}\right) = -RT_C \ln\left(\frac{V_3}{V_4}\right)$.
4. **Adiabatic Compression ($4 \to 1$ from T_C to T_H):** Heat $q = 0$. Work done $w_4 = C_v(T_H - T_C)$.

The net work output per complete cycle is:

$$w_{\text{net}} = q_H + q_C = R(T_H - T_C) \ln\left(\frac{V_2}{V_1}\right)$$

The thermal efficiency η is the ratio of work extracted to heat input:

$$\eta = \frac{w_{\text{net}}}{q_H} = \frac{T_H - T_C}{T_H} = 1 - \left(\frac{T_C}{T_H}\right)$$

The Second Law of Thermodynamics Statements:

- **Kelvin-Planck Statement:** It is impossible to construct a cyclic heat engine that operates solely by absorbing heat from a single reservoir and producing an equivalent amount of work without rejecting heat to a cold sink.
- **Clausius Statement:** It is impossible to construct a cyclic machine whose sole effect is the transfer of heat from a body at a lower temperature to a body at a higher temperature.

2.4 The Third Law of Thermodynamics & Gibbs-Helmholtz Equation

The **Third Law of Thermodynamics** states that the entropy of a pure, perfectly crystalline substance at absolute zero ($T = 0$ K) is exactly zero ($S_0 = 0$).

Gibbs free energy G combines enthalpy and entropy: $G = H - TS$. Since $dG = VdP - SdT$, at constant pressure ($dP = 0$):

$$\left[\frac{\partial G}{\partial T}\right]_P = -S$$

Substituting $-S$ into $G = H - TS$:

$$G = H + T\left[\frac{\partial G}{\partial T}\right]_P$$

For any chemical or physical transformation ($\Delta G = G_{\text{products}} - G_{\text{reactants}}$):

$$\Delta G = \Delta H + T\left[\frac{\partial (\Delta G)}{\partial T}\right]_P$$

This is the celebrated **Gibbs-Helmholtz Equation**. Dividing by T^2 yields the differential form:

$$\left[\frac{\partial \left(\Delta \frac{G}{T}\right)}{\partial T}\right]_P = -\frac{\Delta H}{T^2}$$

3 Chemical Kinetics: Rate Laws & Collision Theory

3.1 Rate of Reaction, Order & Integral Rate Laws

For a general reaction $aA + bB \rightarrow$ Products, the differential rate is:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = k[A]^m[B]^n$$

The overall order of reaction is $n_{\text{total}} = m + n$.

- **First-Order Kinetics:** $-\frac{d[A]}{dt} = k_1[A]$. Integrating from $t = 0$ ($[A] = [A]_0 = a$) to time t ($[A] = a - x$):

$$-\int_a^{a-x} \frac{d[A]}{[A]} = k_1 \int_0^t dt \Rightarrow \ln\left(\frac{a}{a-x}\right) = k_1 t \Rightarrow k_1 = \frac{2.303}{t}\log_{10}\left(\frac{a}{a-x}\right)$$

  The half-life $t_{\frac{1}{2}}$ (time when $x = \frac{a}{2}$) is:

$$t_{\frac{1}{2}} = \frac{\ln 2}{k_1} = \frac{0.693}{k_1}$$

- **Second-Order Kinetics ($2A \rightarrow$ Products):**

$$-\frac{d[A]}{dt} = k_2[A]^2 \Rightarrow \frac{1}{a-x} - \frac{1}{a} = k_2 t \Rightarrow k_2 = \frac{1}{t}\left(\frac{x}{a(a-x)}\right)$$

  The half-life is $t_{\frac{1}{2}} = \frac{1}{k_2 a}$.

3.2 Collision Theory & Arrhenius Equation

According to the Collision Theory of bimolecular gas reactions:

$$\text{Rate} = Z_{AB} e^{-\frac{E_a}{RT}} P$$

where Z_{AB} is collision frequency, $e^{-\frac{E_a}{RT}}$ is the fraction of molecules with kinetic energy exceeding activation energy E_a, and P is the steric orientation factor. The Arrhenius equation describes the exponential increase of rate constant k with absolute temperature T:

$$k = Ae^{-\frac{E_a}{RT}}$$

Taking logarithms:

$$\ln k = \ln A - \frac{E_a}{RT} \quad \Longleftrightarrow \quad \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

Between two temperatures T_1 and T_2:

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

4 Worked VIT Question Bank & Past FAT Problems

Problem 1.1 (VIT FAT Classic - 10 Marks): A Carnot heat engine absorbs 600 kJ of heat from a reservoir at 400° C and rejects heat to a sink at 30° C. Calculate: (i) the Carnot efficiency, (ii) work done per cycle, and (iii) the heat rejected to the sink.

Solution:

1. Convert operating temperatures to Kelvin: $T_H = 400 + 273.15 = 673.15$ K, $T_C = 30 + 273.15 = 303.15$ K
2. Carnot efficiency:

$$\eta = \frac{T_H - T_C}{T_H} = \frac{673.15 - 303.15}{673.15} = \frac{370}{673.15} = 0.5496 \quad (54.96\%)$$

3. Work output per cycle:

$$w_{\text{net}} = \eta \times q_H = 0.5496 \times 600 \text{ kJ} = 329.76 \text{ kJ}$$

4. Heat discharged to the sink:

$$q_C = q_H - w_{\text{net}} = 600 - 329.76 = 270.24 \text{ kJ}$$

Problem 1.2 (VIT Question Bank - 10 Marks): A first-order decomposition reaction has rate constants $k_1 = 3.2 \times 10^{-4}\text{s}^{-1}$ at 25° C and $k_2 = 1.6 \times 10^{-3}\text{s}^{-1}$ at 45° C. Calculate the activation energy E_a and the frequency factor A. ($R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$).

Solution:

1. Convert temperatures: $T_1 = 298.15$ K, $T_2 = 318.15$ K.
2. Ratio of rate constants:

$$\frac{k_2}{k_1} = \frac{1.6 \times 10^{-3}}{3.2 \times 10^{-4}} = 5.0$$

$$\ln(5.0) = 1.6094$$

3. Temperature difference factor:

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{318.15 - 298.15}{298.15 \times 318.15} = \frac{20}{94856.4} = 2.1085 \times 10^{-4}\text{K}^{-1}$$

4. Compute activation energy E_a:

$$E_a = \frac{R \ln\left(\frac{k_2}{k_1}\right)}{\frac{1}{T_1} - \frac{1}{T_2}} = \frac{8.314 \times 1.6094}{2.1085 \times 10^{-4}} = 63462 \text{ J/mol} = 63.46 \text{ kJ/mol}$$

5. Compute Arrhenius pre-exponential factor A:

$$\ln A = \ln k_1 + \frac{E_a}{R T_1} = \ln(3.2 \times 10^{-4}) + \frac{63462}{8.314 \times 298.15} = -8.047 + 25.602 = 17.555$$

$$A = e^{17.555} = 4.20 \times 10^{7}\text{s}^{-1}$$